evaluate_pam50_fusion

August 8, 2026

1 WSI + Arm-Level CNV Fusion for PAM50 Subtyping of TCGA-BRCA and CPTAC-BRCA

1.1 Problem Definition

We address 4-class **PAM50 molecular subtype classification** – the same task as the WSI-only report (`evaluate_pam50`), now with a second modality: arm-level copy-number variation (CNV). Slides labelled *Normal-like* are excluded throughout.

Class	Description
Luminal A	Hormone receptor-positive, low proliferation
Luminal B	Hormone receptor-positive, higher proliferation
Basal-like	Triple-negative, aggressive
HER2-enriched	HER2 overexpression

Two cohorts are in play:

- **TCGA-BRCA** – the training cohort. Three case counts recur below and are not interchangeable: **945** non-Normal cases with arm-level CNV form the CNV arm's fitting set; **910** of them also have WSI features and a fold assignment; **599** have CLAM out-of-fold predictions and are the set every internal head-to-head is computed on.
- **CPTAC-BRCA** – the external cohort: **114 cases** (378 slides). Nothing was ever fitted, tuned or thresholded on it; the fusion rule was fixed on TCGA before the external set was scored. Anything computed on CPTAC after that point is labelled *post hoc* where it appears.

1.2 Pipeline

1.2.1 The WSI branch

Identical to the WSI-only report: each slide is tessellated into non-overlapping patches at 20x magnification, embedded by **UNI2-h** (1536-dimensional per patch), and classified by **CLAM-MB** (`--model_size big`), trained with 10-fold cross-validation on TCGA. On CPTAC the ten TCGA fold models are applied frozen to each slide, their probabilities averaged, and slide-level probabilities mean-pooled per case.

1.2.2 The CNV branch

39 chromosome arms, each the *median* gene-level log2 copy-number over that arm; acrocentric p-arms (13p, 14p, 15p, 21p, 22p) are excluded by construction. The classifier is deliberately simple: `StandardScaler` into a multinomial `LogisticRegression(max_iter=4000, C=0.1, class_weight='balanced')`, imported from `tools/pam50_arms.py` so this notebook and the pipeline scripts describe the same model by construction.

Arm-level rather than focal, because shallow whole-genome sequencing resolves arm and segment scale: the assay is cheap and clinically reachable, whereas a model over the 19,755 focal GISTIC calls in the same download would not transfer to an sWGS setting. CNV rather than RNA, because the PAM50 labels are computed from the expression matrix itself (`project/data/pam50.R`) – an RNA branch predicting PAM50 leaks the target by construction, and copy number is a different assay with no such circularity.

1.2.3 The fusion rule

The equal-weight mean of the two probability vectors, `(P_wsi + P_cnv) / 2`. It is untrained – one line of arithmetic, nothing fitted beyond the two unimodal arms.

1.3 Experimental Setup

Both arms are evaluated on TCGA under `splits/tcga_brca_subtyping_100` (k = 10, seed 1) – the split set of the frozen WSI baseline `pam50_final_s1`, reused unchanged so the WSI-only column below is that exact model, not a retrain.

One property of this split set has to be stated plainly, and it is where this report corrects its predecessor: **CLAM's 10 splits are drawn independently, not partitioned.** 599 of the 910 eligible cases land in at least one test fold and 242 land in two to five (both counts recomputed below), so "WSI only" is in effect a small ensemble of up to five fold models per case – an audit found this flatters it by roughly +0.01 AUROC, and re-running with a random stratified partition gave the same verdict. The WSI-only report's claim (its section 1.3.3) that each slide appears in exactly one test fold does not hold for this split set.

All confidence intervals are paired bootstraps: every model is scored on the same resamples of the same cases, so each contrast is a difference over shared draws. The head-to-head tables use 4,000 resamples at seed 7 (constants imported from `tools/evaluate_cnv_wsi_fusion.py`); the operator ladder uses 2,000 at seed 13, mirroring `tools/compare_fusion_ladder.py`. The protocol behind every number is stated where the number appears.

```
helpers resolve to: /workspace/dp-code/tools/pam50_arms.py
CNV arm: StandardScaler -> LogisticRegression(max_iter=4000,
           C=0.1, class_weight='balanced')
bootstrap: 4000 resamples, seed 7 | internal CNV CV: 10-fold, seed 0
ladder bootstrap: 2000 resamples, seed 13
```

1.4 Results

1.4.1 Inputs, as read from disk

Every number below is computed in this notebook from:

- `.scratch/results/pam50_final_s1/split_{0..9}_results.pkl` – the WSI arm, out-of-fold
- `.scratch/cptac_validation/.../ensemble_predictions.csv` – the WSI arm on CPTAC
- `.datasets/cnv/{tcga,cptac}_brca_cna_arm.csv` – arm-level CNV, both cohorts

- `.scratch/results/pam50_wsi_cnv_*_s1/` – the five trained ladder runs

Two class orders are in play in those files (CLAM's `LumA, LumB, Basal, Her2` against `pam50_arms.CLASSES`'s sorted `Basal, Her2, LumA, LumB`); the recovered mapping is printed and asserted to be a permutation before anything is scored.

Table 1: Cohort inventory -- three TCGA counts, none interchangeable

case set	n	LumA	LumB	Basal	Her2
TCGA: CNV fitting set (non-Normal, arm CNV)	945	499	197	171	78
TCGA: CLAM tabular table (CNV + WSI + fold)	910	475	195	165	75
TCGA: with CLAM out-of-fold predictions	599	318	124	106	51
CPTAC: external cases	114	56	17	27	14

```
  CPTAC probabilities read from 378 slide rows, mean-pooled per case
  split multiplicity: 357 cases tested by exactly 1 fold, 242 by 2-5
  CLAM class order recovered: ['LumA', 'LumB', 'Basal', 'Her2']
  scoring order (CLASSES):    ['Basal', 'Her2', 'LumA', 'LumB']
```

1.4.2 Internal TCGA head-to-head – 599 cases, both arms out-of-fold

WSI probabilities are `pam50_final_s1`'s pooled out-of-fold predictions. The CNV arm is refit per CLAM fold – trained on that fold's training cases, applied to its held-out cases – so **both arms are out-of-fold on the same folds of the same 599 cases**; of the three defensible internal CNV protocols (all measured in the next table), that matching is why this one is the figure the report carries. Paired bootstrap, 4,000 resamples, seed 7. The recall columns are counts at argmax.

Table 2: Internal TCGA head-to-head (599 cases, CNV refit per CLAM fold)

model	AUROC	95% CI	balAcc	LumA	LumB	Basal	Her2
WSI	0.887	[0.865, 0.908]	0.677	257/318	66/124	91/106	26/51
CNV	0.872	[0.848, 0.895]	0.678	231/318	78/124	92/106	25/51
Mean	0.926	[0.909, 0.941]	0.751	263/318	79/124	101/106	30/51

contrast	dAUROC [95% CI]		dBalAcc [95% CI]	
Mean - WSI	+0.039 [+0.024,+0.054]	sig	+0.074 [+0.028,+0.122]	sig
Mean - CNV	+0.054 [+0.036,+0.072]	sig	+0.073 [+0.031,+0.116]	sig
WSI - CNV	+0.015 [-0.014,+0.045]	ns	-0.001 [-0.065,+0.065]	ns

1.4.3 One internal CNV figure, three defensible protocols

Three internal CNV-only figures circulate in this project's documents, and the spread between them is larger than contrasts this report calls significant, so the protocol is pinned rather than left to the reader: the per-CLAM-fold refit above is the nominated figure **because it is the only protocol under which both arms are out-of-fold on the same folds**. The other two are recomputed here so any previously seen number can be placed.

Table 3: The internal CNV-only macro AUROC under its three protocols

0.8721	refit per CLAM fold, 599 cases	<< carried here
0.8624	StratifiedKFold(10, seed 0), 599 cases	
0.8657 +- 0.0034	5-fold CV x 10 seeds (0-9), 945 cases	<< published

1.4.4 External validation on CPTAC – the headline table

The CNV arm is fit once on all 945 TCGA cases and applied to CPTAC unchanged; the WSI arm's CPTAC probabilities are read from the frozen inference run. The first three rows were specified before CPTAC was scored. The fourth was not: the prior-balanced mean divides the WSI probabilities by the TCGA training prior before averaging (the CNV arm is class-balanced by construction, the WSI arm is not) and was run **post hoc**, after the raw fusion underperformed on balanced accuracy – it is a decision-rule control, not the pre-registered analysis. The headline contrast is therefore fusion against raw WSI, marked below; fusion against CNV alone is printed beside it every time, because that margin is the marginal one.

```
external set: 114 cases
TCGA training prior: LumA 0.528  LumB 0.208  Basal 0.181  Her2 0.083
```

Table 4: External CPTAC -- TCGA-trained, nothing refit or tuned on CPTAC

model	AUROC	95% CI	balAcc	LumA	LumB	Basal	Her2
WSI	0.847	[0.793, 0.897]	0.513	52/56	4/17	24/27	0/14
CNV	0.888	[0.833, 0.935]	0.716	37/56	9/17	22/27	12/14
Mean	0.909	[0.861, 0.949]	0.646	50/56	7/17	23/27	6/14
Mean(bal) post hoc	0.912	[0.864, 0.952]	0.740	43/56	10/17	24/27	10/14

contrast	dAUROC [95% CI]		dBalAcc [95% CI]	
Mean - WSI	+0.063 [+0.024,+0.104]	sig	+0.133 [+0.031,+0.240]	sig *
Mean - CNV	+0.021 [-0.003,+0.047]	ns	-0.069 [-0.165,+0.022]	ns
CNV - WSI	+0.042 [-0.019,+0.103]	ns	+0.202 [+0.089,+0.315]	sig
Mean(bal) - Mean	+0.003 [-0.005,+0.011]	ns	+0.093 [+0.014,+0.177]	sig
Mean(bal) - CNV	+0.024 [+0.000,+0.050]	sig	+0.024 [-0.055,+0.099]	ns

* the pre-specified headline contrast; Mean(bal) rows are post hoc

```
Her2 rescue control (WSI arm): raw 0/14, prior-balanced (12x) 0/14,
  SLD-EM 0/14 (estimated Her2 prior 0.000 against a true 0.123)
error correlation phi(WSI, CNV), external, independent arms: -0.006
```

1.4.5 Per-class external performance

Per-class one-vs-rest AUROC and recall at argmax on the same 114 cases. Her2 is where the modalities part company: by *ranking* the two arms are comparable, but the WSI arm converts none of it into calls – and the rescue control above shows no monotone prior reweighting changes that, so it is not a threshold artifact. The class supports (Her2 n = 14, LumB n = 17) are printed with the table; at that n these per-class figures are indicative, not precise.

Table 5: External per-class AUROC (one-vs-rest) and recall at argmax

one-vs-rest AUROC:

model	LumA	LumB	Basal	Her2
WSI	0.861	0.693	0.972	0.860
CNV	0.883	0.827	0.972	0.871
Mean	0.916	0.848	0.992	0.881
Mean(bal) post hoc	0.918	0.848	0.990	0.893

recall at argmax:

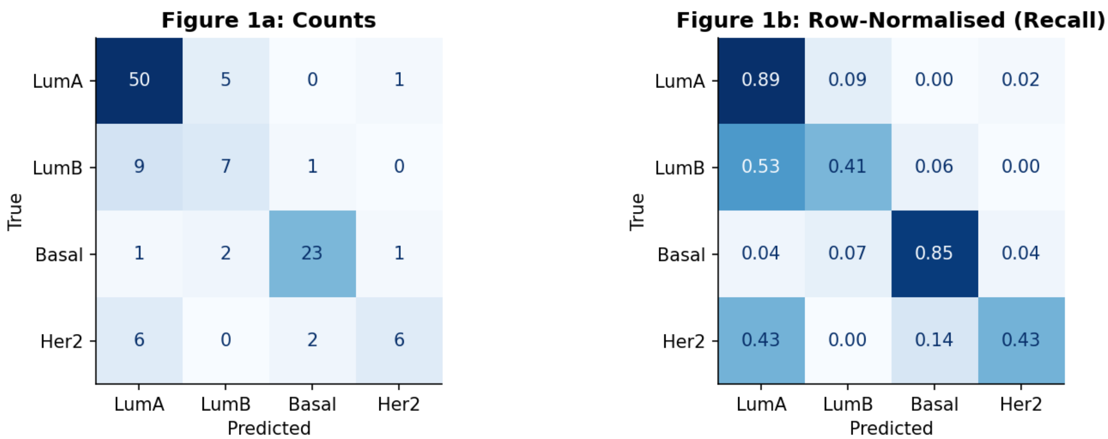
model	LumA	LumB	Basal	Her2
WSI	52/56	4/17	24/27	0/14
CNV	37/56	9/17	22/27	12/14
Mean	50/56	7/17	23/27	6/14
Mean(bal) post hoc	43/56	10/17	24/27	10/14

class support: LumA n=56 LumB n=17 Basal n=27 Her2 n=14

1.4.6 Confusion matrix, external equal-weight mean

Figure 1 shows where the fused model's external decisions land, as raw counts (left) and row-normalised recall (right). The Her2 row is the one to read: the fusion keeps 6 of 14 under the raw WSI decision rule, against the CNV arm's 12 of 14 (Table 5).

Figure 1: External CPTAC Confusion Matrix -- Equal-Weight Mean (TCGA-trained, nothing refit; n = 114 cases)



1.4.7 ROC curves, external equal-weight mean

Figure 2 shows one-vs-rest ROC curves for the equal-weight mean on the 114 external cases. Ranking survives transfer for every class – including Her2, whose curve sits near the CNV arm's despite the argmax collapse of the WSI branch.

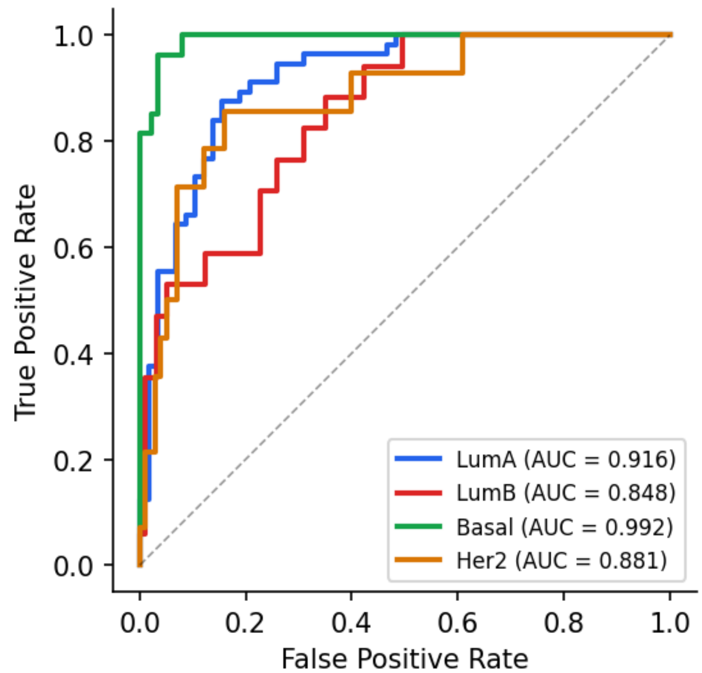


Figure 2: One-vs-Rest ROC -- External CPTAC, Equal-Weight Mean (n = 114)

1.4.8 The fusion-operator ladder – internal only

Five trained fusion operators – concat, gated, cross_attention, film_attention, coattn – were run on the same splits (CLAM-MB, the WSI branch warm-started from `pam50_final_s1` and not frozen, k = 10, seed 1). They are compared as pooled out-of-fold predictions over the 599 cases all runs share, with a 2,000-resample paired bootstrap at seed 13, mirroring `tools/compare_fusion_ladder.py`. **The baseline each operator is read against is the equal-weight mean, not the WSI-only model.**

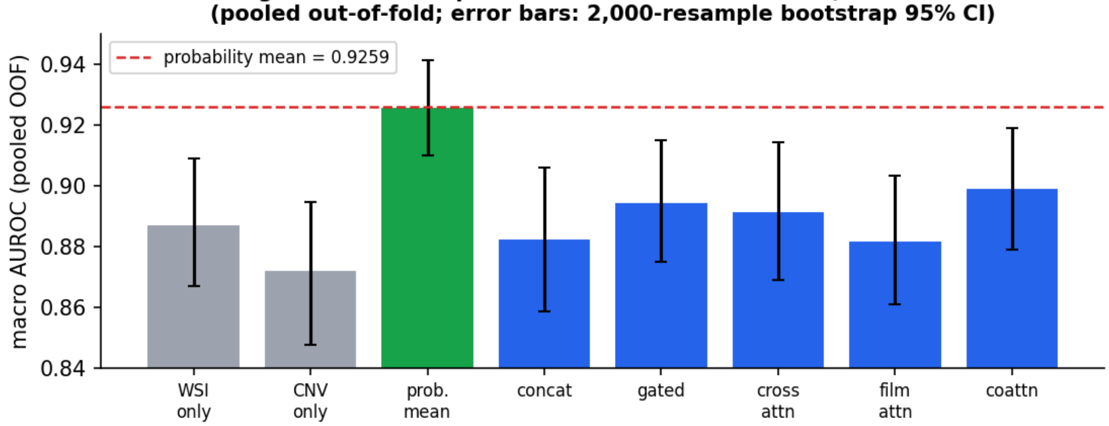
Where a reader would expect an external CPTAC column for these five rows: there is none, and none can currently be produced. `project/CLAM/evaluate_multimodal.py` has no loading path for `film_attention` or `coattn` checkpoints (the former raises in fusion-mode inference, the latter is misidentified as `cross_attention` and fails `load_state_dict(strict=True)`), so the ladder is an internal-only result by construction and the external table above keeps exactly four model rows.

The mechanism check under the table contrasts two error-correlation regimes, both internal and pooled out-of-fold: phi among the five jointly trained operators against phi between the two independently trained unimodal arms (CNV per-CLAM-fold refit). Models sharing a trunk and an initialisation make the same mistakes, so ensembling them recovers little; the two independent arms err nearly independently, which is the signal the untrained mean exploits.

```
Table 6: Fusion-operator ladder, pooled OOF (599 internal cases)
============================================================================
            arm  AUROC         95% CI balAcc                dAUROC vs mean
       WSI only 0.8872 [0.867, 0.909] 0.6772 -0.0386 [-0.0544,-0.0234] sig
       CNV only 0.8721 [0.848, 0.895] 0.6784 -0.0541 [-0.0732,-0.0371] sig
probability mean 0.9259 [0.910, 0.941] 0.7513            --  (the baseline)
         concat 0.8827 [0.859, 0.906] 0.6741 -0.0430 [-0.0630,-0.0256] sig
          gated 0.8947 [0.875, 0.915] 0.6832 -0.0311 [-0.0477,-0.0161] sig
 cross_attention 0.8917 [0.869, 0.914] 0.6848 -0.0342 [-0.0541,-0.0156] sig
  film_attention 0.8818 [0.861, 0.903] 0.6652 -0.0438 [-0.0618,-0.0271] sig
         coattn 0.8992 [0.879, 0.919] 0.6842 -0.0267 [-0.0432,-0.0106] sig
----------------------------------------------------------------------------
  dBalAcc vs mean: 7/7 arms significant,
      deltas -0.086 to -0.066 (all below the mean)

error correlation phi -- each value labelled with its regime:
  5 jointly trained operators, internal pooled OOF, 10 pairs: mean 0.656
      (min 0.582, max 0.706)
  WSI-only vs CNV-only, internal, CNV per-CLAM-fold refit:    0.193
  WSI-only vs CNV-only, external CPTAC (from Table 4's arms): -0.006
============================================================================
```

Figure 3: Fusion-Operator Ladder -- Internal TCGA, 599 Cases
(pooled out-of-fold; error bars: 2,000-resample bootstrap 95% CI)



1.5 Summary

The closing block gathers the four external rows, the internal comparison, and the verdict.

```
=======================================================================
  WSI + arm-level CNV | 4-class PAM50 | trained on TCGA, external CPTAC
=======================================================================
--- External CPTAC, n = 114 cases (4000-resample paired bootstrap) ---
  WSI                    AUROC 0.847 [0.793, 0.897]  bal 0.513  Her2  0/14
```

```
  CNV                  AUROC 0.888 [0.833, 0.935]  bal 0.716  Her2 12/14
  Mean                 AUROC 0.909 [0.861, 0.949]  bal 0.646  Her2  6/14
  Mean(bal) post hoc   AUROC 0.912 [0.864, 0.952]  bal 0.740  Her2 10/14

--- Per-class, equal-weight mean (external) ---
  Class    AUROC   Recall
  LumA     0.916    50/56
  LumB     0.848     7/17
  Basal    0.992    23/27
  Her2     0.881     6/14

--- Internal TCGA, 599 cases (CNV refit per CLAM fold) ---
  WSI 0.8872   CNV 0.8721   equal-weight mean 0.9259
  trained operators (5): 0.8818 - 0.8992, all significantly below
  the mean (internal only; no external evaluation path exists)

VERDICT: conditioning H&E on copy number does not beat averaging two
independently trained predictions. The equal-weight mean is the best model
internally and externally; every trained operator sits significantly below
it. Fusion's edge over CNV alone is marginal -- report both (Table 4).
========================================================================
```

1.6 Limitations

The external Her2 collapse is the report's most consequential finding and its clearest limitation. The WSI arm calls 0 of 14 Her2 cases on CPTAC while calling 26 of 51 internally (Table 2), and the calibration explanation was tested and refuted in the rescue control under Table 4: recall stays 0/14 after dividing by the training prior (a roughly 12x Her2 boost) and under unsupervised SLD-EM prior estimation, which drives the implied Her2 prior to zero against a true 0.123. The collapse is induced by the domain shift, so remediation belongs on the imaging side – stain normalisation, encoder choice – not in recalibration.

Power is thin exactly where the story is. Her2 has 14 external cases and LumB 17, so per-class external estimates are indicative rather than precise, and the 12-of-14 against 0-of-14 contrast, however striking, rests on 14 cases.

The two cohorts' features are geometrically comparable but not identically produced: the CPTAC `.h5` files carry CLAM `create_patches_fp` attributes while TCGA's carry Trident ones, so tissue segmentation – and therefore which tiles enter each bag – differs between cohorts. That difference is a live confound inside the domain-shift explanation above.

The ladder result carries one open confound of its own. All five operators warm-start their WSI branch from the same `pam50_final_s1` checkpoint, so "joint training collapses diversity" and "shared initialisation collapses diversity" are not yet separated; a from-scratch arm would settle it. Finally, run-to-run variance has never been measured for these runs – seeds and folds are fixed, no tolerance exists, the attention-based operators are not bitwise reproducible – so every interval here comes from the stated bootstrap or seed spread; calibration, which no arm measures, is left out rather than estimated.